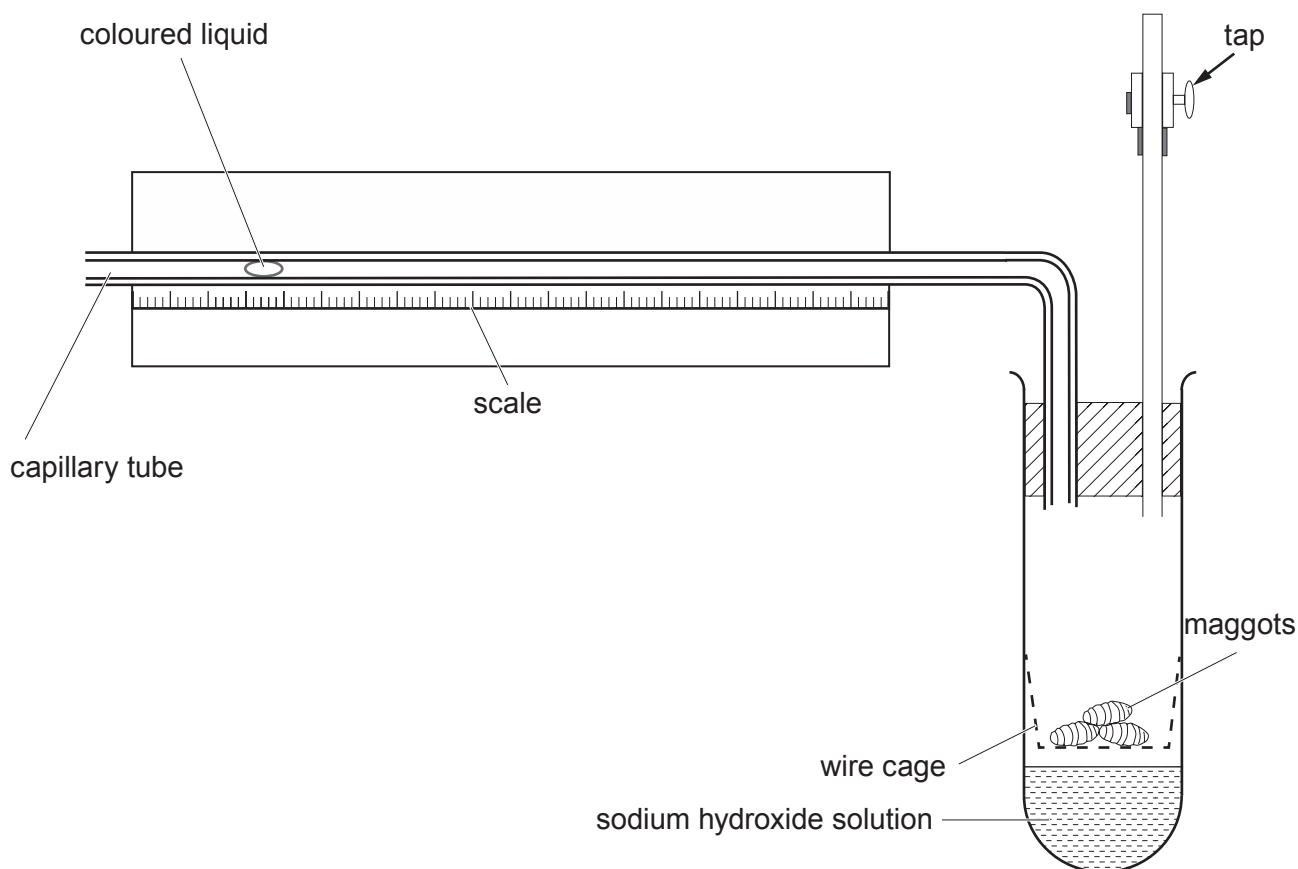


5. A class of 20 students investigated respiration in maggots over a 5 minute time period. The diagram below shows the apparatus that they used.



In experiment 1:

- 5 cm^3 of sodium hydroxide solution was placed into the test tube.
- Ten maggots were placed into the wire cage.
- The tap was left open.
- The test tube containing the maggots was placed in a water bath at 25°C .
- After 10 minutes, the tap was closed and the position of the coloured liquid noted.
- The apparatus was left for five minutes and the position of the coloured liquid was noted at one minute intervals.

Sodium hydroxide absorbs carbon dioxide and is corrosive.

In experiment 2, the same procedure was carried out using 5 cm^3 of water instead of sodium hydroxide solution.

- (a) (i) Suggest why the maggots were placed in the wire cage.

[1]

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(ii) Give **two** reasons why the apparatus was left in the water bath with the tap open for ten minutes before readings were taken. [2]

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(iii) Suggest a suitable control for these experiments. Give a reason for your answer. [2]

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Each student carried out both experiments and the class results were combined. The mean results are shown in the table below.

Time / minutes	Mean position of coloured liquid / mm	
	Experiment 1 (with sodium hydroxide)	Experiment 2 (with water)
0	0	0
1	13	0
2	22	0
3	35	0
4	49	0
5	61	0

(b) (i) Explain why the coloured liquid moved in experiment 1 but did not move in experiment 2. [3]

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(ii) Explain the advantages of collecting results from the **whole** class. [2]

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(iii) In these experiments the diameter of the lumen of the capillary tube was 1.2 mm and the total mass of maggots was 0.5 g.
Calculate the rate of oxygen consumption per g for the maggots during the 5 minutes of the experiment. Give the answer in mm³ g⁻¹ minute⁻¹ to 1 decimal place. [3]

The formula for the volume of a cylinder = $\pi r^2 h$

$\pi = 3.14$

rate of oxygen consumption = mm³ g⁻¹ minute⁻¹



- (c) Scientists can use the volume of carbon dioxide produced and the volume of oxygen consumed in a given time to give the respiratory quotient (RQ).
The equation for working out the RQ is:

$$\text{RQ} = \frac{\text{number of molecules of CO}_2 \text{ produced}}{\text{number of molecules of O}_2 \text{ consumed}}$$

The equation for respiration of glucose is:



As the equation shows, 6 molecules of CO₂ are produced and 6 molecules of O₂ are consumed when 1 molecule of glucose is respired.

The RQ value for glucose is 1.

The equation for respiration of the fat tripalmitin is:



- (i) Calculate the RQ for tripalmitin. [1]

RQ =

- (ii) Use the equation for respiration of tripalmitin to suggest why desert animals such as camels use fat as a substrate for respiration. [2]

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(iii) The respiration of fat releases more energy than the respiration of glucose produced by the breakdown of glycogen.

Explain why muscles use glycogen as an energy store rather than fat. [2]

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